Catching Denial-of-Service Attacks With an AI Threat Detector

A Random Forest ensemble trained on NSL-KDD network traffic: what it catches, what it misses, and how a 2025 company should use it

Ruthvik Nath Bandari

AAI6680: AI for Cybersecurity · Northeastern University · Mimoza Dimodugno, PhD

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The threat: denial-of-service floods that take businesses offline

- **What it is:** a flood of traffic that overwhelms a server so real customers cannot get in.
- **Scale:** Cloudflare blocked **47.1 million** DDoS attacks in 2025, more than double 2024; the largest reached **31.4 Tbps**.
- **Financial impact:** an hour of downtime costs **over $300,000** for more than 90% of mid-size and large firms.
- **In the data:** DoS is **36%** of NSL-KDD training traffic, the largest attack family.

RECENT EXAMPLE

Threema, August 2026. DDoS attacks left the messaging service unavailable or only partly available for an evening and the next morning, and also targeted its colocation partner, Nine.

Sources: Yoachimik et al. (2026); DiDio (2024); Ilascu (2026); Tavallaee et al. (2009).

APPROACH

Applying Chapter 5: network anomaly detection with AI

1 Load

125,973 labelled connections (NSL-KDD)

2 Clean

No missing or duplicate rows; constant column dropped; 121 features

3 Train

Random Forest: 200 decision trees vote on each connection

4 Test

80/20 split, then the official test file with new attacks

TWO CHAPTER 5 APPROACHES

- **Supervised classification** (this project): learns labelled attacks; excellent on known threats.
- **Anomaly detection** (Modules 1 and 3): learns normal traffic only; flags anything unusual, including new attacks.

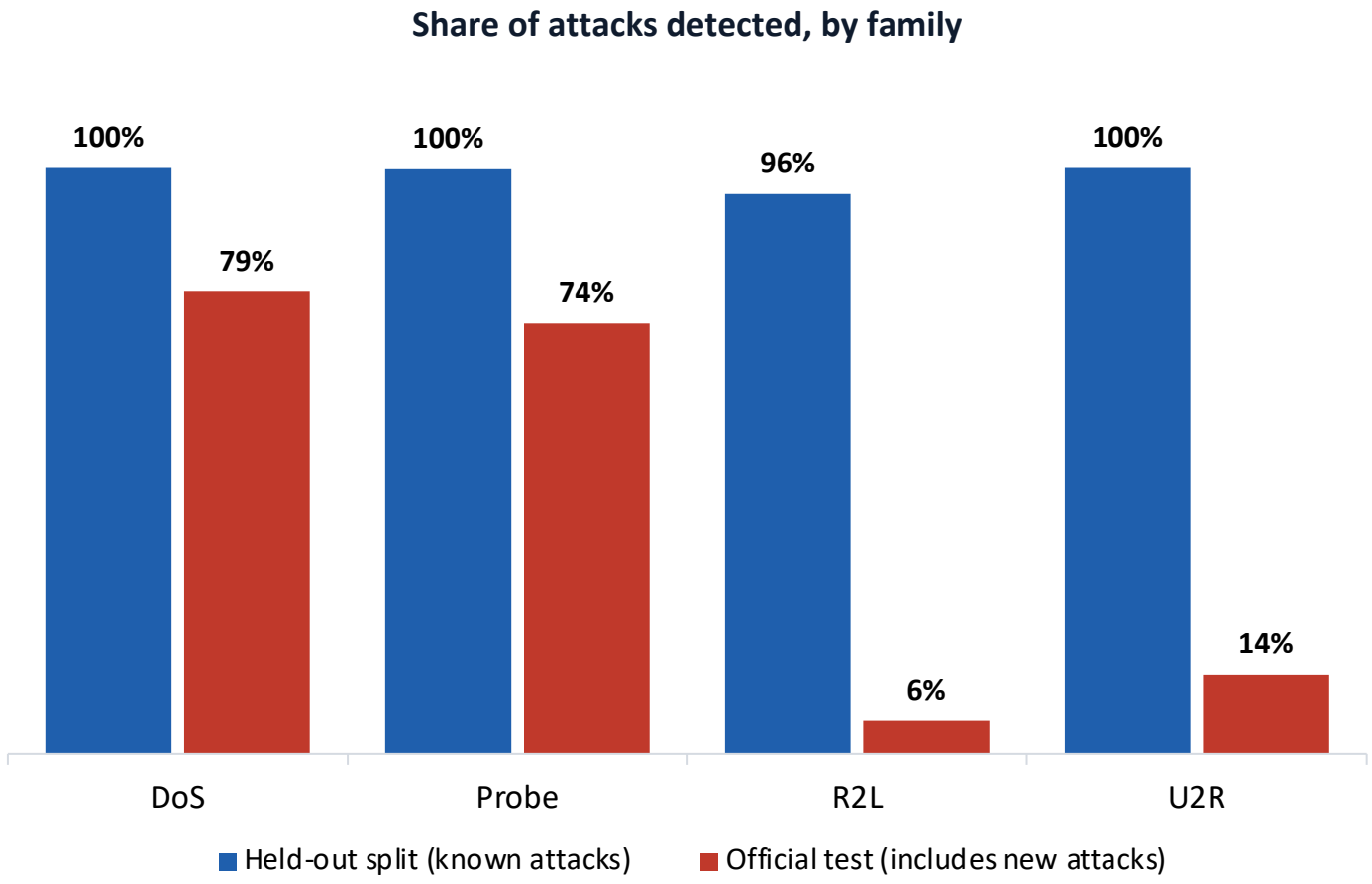
WHY A RANDOM FOREST

- An **ensemble** of many trees is more accurate and stable than one tree, and it shows **which features** drive decisions.

Sources: Parisi (2019, Chapter 5); Tavallaee et al. (2009).

99.90% accurate on familiar traffic, 76.7% with new attacks

- **Held-out split:** 99.90% accuracy, 7 false alarms in 13,469 normal connections.
- **Official test set** (17 unseen attack types): 76.7% accuracy overall.
- **New attack types** caught only 23% of the time, against 77% for known ones.



Source: author's analysis of NSL-KDD (Tavallaee et al., 2009). Held-out U2R bar rests on only 6 connections.

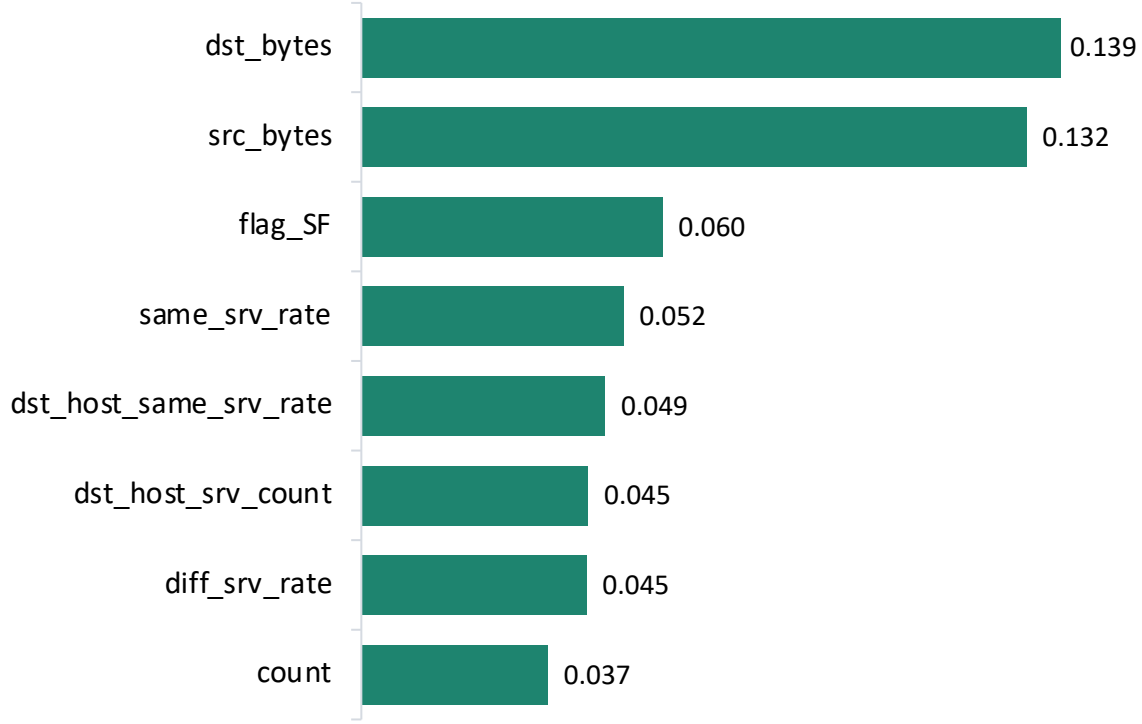
Why floods are easy to catch: they look nothing like normal traffic

DOS VS NORMAL (MEDIAN PER CONNECTION)

	DoS	Normal
Connections to same host in 2 s	**172**	4
Share with SYN errors	**100%**	0%
Share to the same service	**7%**	100%
Bytes sent by source	**0**	233

DoS was detected 78.9% of the time on the official test set, the best of any family.

What the Random Forest relies on (importance)



Importance across all attacks: byte counts, connection status and traffic patterns separate floods.

Sources: author's analysis; Parisi (2019, Chapter 5).

TOP THREATS AND LIMITS

The real danger: the attacks the detector misses

MOST-MISSED ATTACK TYPES (OFFICIAL TEST SET)

Attack	Family	Missed	
guess_passwd	R2L	1,231	known
warezmaster	R2L	795	known
processtable	DoS	675	new
mscan	Probe	636	new
apache2	DoS	588	new
snmpguess	R2L	331	new

WHAT THIS MEANS

- **New DoS variants** (processtable, apache2) slip past: the model never saw them in training.
- **Quieter attacks** such as password guessing are missed too: only 5.6% of R2L attacks were caught.
- **Mitigation (Chapter 5):** add anomaly detection, which flags unusual traffic it was never shown.

Sources: author's analysis; Parisi (2019, Chapter 5).

Trend: autonomous DDoS mitigation, and where this detector fits

TREND: AUTONOMOUS DDOS MITIGATION

Cloudflare blocked about **5,343 network-layer DDoS attacks per hour** in the first half of 2026. Its systems act without human intervention, because attacks end before an alert would reach an analyst.

HOW A COMPANY SHOULD USE THIS DETECTOR

- **Automate the confident cases:** block clear floods in real time.
- **Route uncertain alerts to analysts:** on new traffic, 2.7% of normal connections were wrongly flagged.
- **Add anomaly detection and retrain** as new attacks appear.

Sources: Cloudforce One (2026); author's analysis.

WHAT I LEARNED

Reflection: accuracy is not the same as protection

My first result, **99.90%**, looked finished. The official test set (**76.7%**, and only **23%** of new attacks caught) showed the model knew the attacks it had seen, not attacks in general.

My Module 1 and 3 anomaly detectors had the opposite profile: supervised and anomaly-based detection cover each other's blind spots.

As a future security engineer, I want to build detectors that are judged on the threats they miss, and that let people make the calls a model is not sure about.

Reflection draws on my Module 1, Module 3 and Module 4 labs.

APA 7TH EDITION

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